

Pods in Kubernetes

Defining Kubernetes Pods



David Clinton

Data Analyst | Linux System Administrator

bootstrap-it.com | linkedin.com/in/dbclinton



```
apiVersion: apps/v1
kind: Deployment
```

A Deployment Definition:

```
metadata:
```

```
  name: my-app
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: my-app
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: my-app
```

```
    spec:
```

```
      containers:
```

```
        - name: app-container
```

```
          image: my-app:latest
```

```
          env:
```

```
            - name: APP_MODE
```

```
              valueFrom:
```

```
                configMapKeyRef:
```

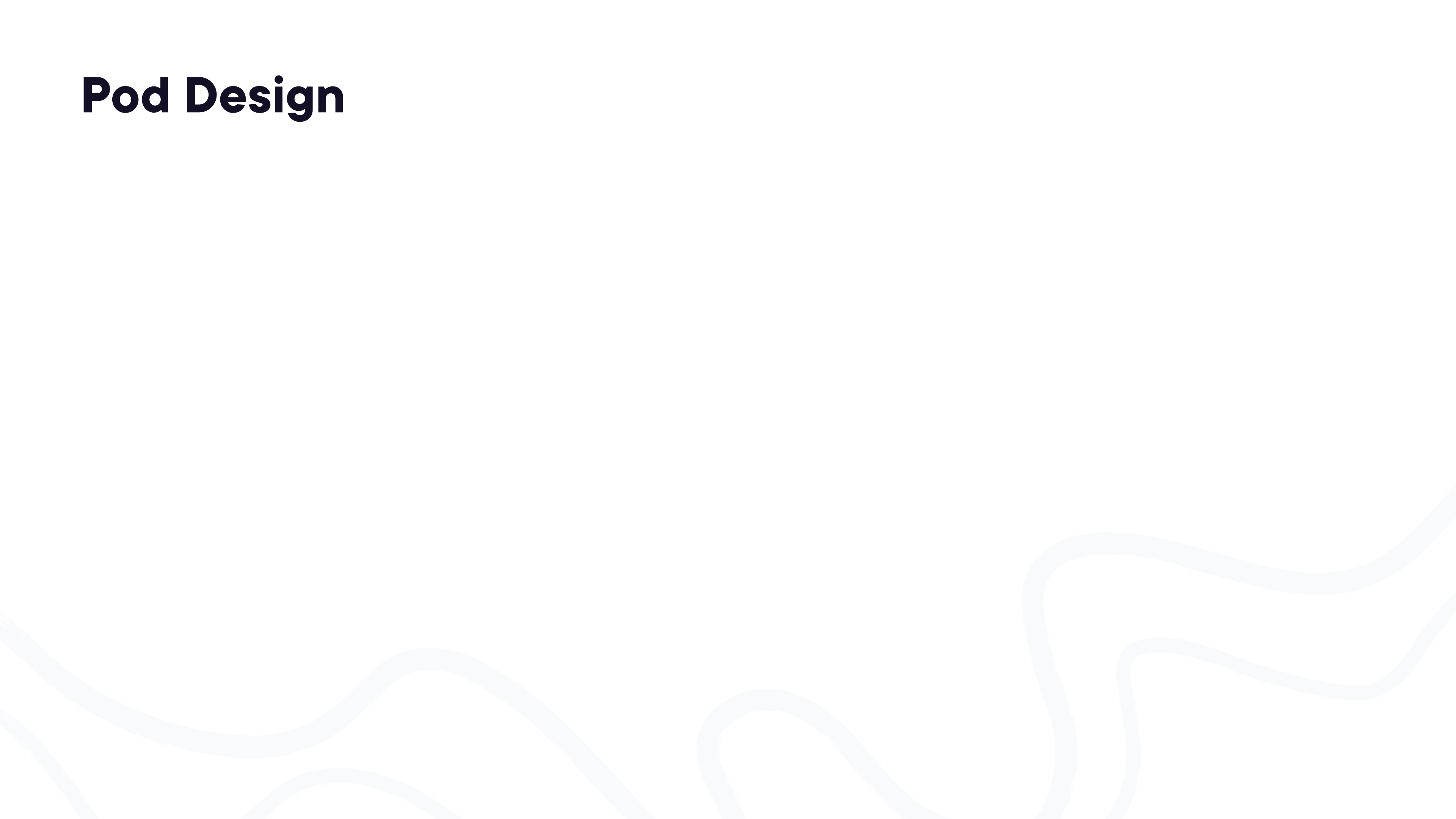
A Service Definition:

```
apiVersion: v1
kind: Service
metadata:
  name: my-app-service
spec:
  selector:
    app: my-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
  type: ClusterIP
```

A ConfigMap Definition:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config
data:
  APP_MODE: "production"
  LOG_LEVEL: "info"
```

Pod Design



Pod Design

Single Container

Pod Design

Single Container

Multiple Container

Pod Design

Single Container

Multiple Container

Ephemeral

Pod Design

Single Container

Multiple Container

Ephemeral

Stateful

Pod Design

Single Container

Multiple Container

Ephemeral

Stateful

Declarative



Designing a Single-container Pod



Designing a Multi-container Pod